

ORIGINAL RESEARCH

Metric-based pill recognition with the help of textual and visual cues

Richárd Rádli | Zsolt Vörösházi  | László Czúni 

Image Processing Research Laboratory, University of Pannonia, Veszprém, Hungary

Correspondence

László Czúni, Image Processing Research Laboratory, University of Pannonia, Veszprém, H-8200, Hungary.
Email: czuni.laszlo@mik.uni-pannon.hu

Funding information

Nvidia; Nemzeti Kutatási, Fejlesztési és Innovációs Alap, Grant/Award Numbers: 2020-1.1.2-PIACI-KFI-2021-00296, OTKA K-135729, TKP2021-NVA-10

Abstract

Pill image recognition by machine vision can reduce the risk of taking the wrong medications, a severe healthcare problem. Automated dispensing machines or home applications both need reliable image processing techniques to compete with the problem of changing viewing conditions, large number of classes, and the similarity in pill appearance. The problem is attacked with a multi-stream, two-phase metric embedding neural model. To enhance the metric learning procedure, dynamic margin setting is introduced into the loss function. Moreover, it is shown that besides the visual features of drug samples, even free text of drug leaflets (processed with a natural language model) can be used to set the value of the margin in the triplet loss and thus increase the recognition accuracy of testing. Thus, besides using the conventional metric learning approach, the given discriminating features can be explicitly injected into the metric model using the NLP of the free text of pill leaflets or descriptors of images of selected pills. The performance on two datasets is analysed and a 1.6% (two-sided) and 2.89% (one-sided) increase in Top-1 accuracy on the CURE dataset is reported compared to existing best results. The inference time on CPU and GPU makes the proposed model suitable for different kinds of applications in medical pill verification; moreover, the approach applies to other areas of object recognition where few-shot problems arise. The proposed high-level feature injection method (into a low-level metric learning model) can also be exploited in other cases, where class features can be well described with textual or visual cues.

1 | INTRODUCTION

The recognition of medicinal pills is a challenging task since high accuracy is required. At the same time, the large number of classes is often paired with nuances in differences in visual appearance and significant impacts in viewing conditions. The task of recognising pills is not only challenging for computer vision systems but also for humans. According to [1], drug errors are the most frequent mistakes in healthcare. A deep dispensing error analysis was described in [2]. It was found that 2.1% of dispensing were reported to have one or more dispensing errors. At the final check stage (inside the hospital pharmacy), 53% of all errors were classified as content errors, while this number was 70% when checked outside of the pharmacy. On a broader work introduced in [3], it is found that wrong drug type (17%) and drug quantity (12%)

were among the most common dispensing error types based on a survey conducted in several hospitals in Wales. Both could be reduced with automatised verification systems. The risks of taking the wrong medications can be so serious worldwide that the WHO has chosen Medication Safety as the theme for the World Patient Safety Day in 2022 (“Medication Without Harm”—<https://www.who.int/campaigns/world-patient-safety-day/2022>). To promote the development of accurate and efficient computer solutions, the United States National Library of Medicine (NLM) announced an algorithm challenge on pill recognition in 2016, [4]. After evaluating the results, the accuracy of the first three submissions did not seem sufficient to develop a mobile online service for matching consumer-quality images.

While many new approaches have emerged, there remains a pressing need to enhance the performance of pill

This is an open access article under the terms of the [Creative Commons Attribution-NonCommercial-NoDerivs](https://creativecommons.org/licenses/by-nc-nd/4.0/) License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.

© 2024 The Author(s). *IET Image Processing* published by John Wiley & Sons Ltd on behalf of The Institution of Engineering and Technology.

identification systems. This is particularly important for their use in less controlled environments, where their current capabilities are limited. Prevented dispensing incidents is crucial for ensuring patient safety and improving the overall quality of healthcare services. Due to the growing number of elderly people, assistive technologies and hospitals could use pill verification systems during manual pill dispensing.

In our article, we deal with pill metric learning, which aims to learn the visual discriminating features of pills that are necessary to differentiate them. Metric learning has an obvious advantage over direct object recognition when recognising new (untrained) pills, which is not a rare case in medical treatment. To enhance the distinguishing abilities of deep neural networks, we implement multi-stream data processing with preprocessed dedicated inputs and fuse the outputs of the different streams. As our experimental ablation models show in Section 4.6, in parallel with previous findings in [5–7], the chosen approach shows significant advantages over single-stream architectures. The design of the proposed streams is uniform and avoids special training steps such as OCR (optical character recognition) of imprints. Since pills are designed with the intention that each type has different visual appearance, high-level visual descriptors (or say rules) can be defined to describe the appearance, namely: colour, shape, size, embossing, carving, imprint, and score. In our approach, we show that besides using the conventional metric learning approach, the named visual features can be explicitly injected into the metric model using the NLP (natural language processing) of the free text of pill leaflets or the descriptor of images of selected pills.

In our paper, as the most significant contribution, we introduce the methods and advantages of the different injection techniques. As a result, the discriminating ability of the metric neural model is improved since we can quantify the degree of visual differences between samples, and not only binary information (“same class”, “different class”) rules the metric embedding process. It is a substantial difference to existing metric learning approaches (such as [6–9]) where only positive or negative examples are given to the model during training, and their differences are not quantified. Thanks to the new design of our multi-stream DNN models and the new injection techniques, besides other advantages, our tests show a 1.6–2.89% increase in Top-1 accuracy on the CURE dataset compared to the best previous models.

In the next section we summarise and discuss related studies, outlining the differences from our framework. Section 3 describes the two datasets we use in the experiments (in Section 3.2), the main steps of our approach: preprocessing (detection and segmentation in Section 3.3), the construction of the data streams (in Section 3.4), and then the alternatives of adding high-level cues (in Section 3.5). The details of the loss function are in Section 3.6. For the experimental validation of our ideas, Section 4 contains the details of different tests on the two datasets, including ablation studies in Section 4.6, results are discussed in Section 5. Finally, Section 6 presents conclusions and future work.

2 | LITERATURE OVERVIEW AND DISCUSSION

Since reliable automatic pill recognition is an important challenge for the industry, there is a large number of attempts, which we group now into three sets:

1. Generic object detectors;
2. Metric learning methods;
3. Multimodal approaches.

Many generic object detectors are applied to similar problems as pill recognition, so it is obvious to check their performance on pills. In [10], a two-stage approach was applied: during the detection stage, a ResNet-50 [11] based deep convolutional neural network was applied to extract features and construct a feature pyramid. Then, still in the first phase, the subsequent regression and classification layers were responsible for determining the position of the pills. In the second stage, based on these positions, the Xception [12] deep convolutional neural network is used to classify the pill types. This approach could reach 79.4% Top-1 accuracy on the authors' dataset. Naturally, handling the problem of few-shot recognition, an essential aspect due to the large number of possible classes, is far from this approach.

Since the large number of pill classes is a costly problem when training recognition systems, the solution in [13] proposes an algorithm which can improve the detection performance based on limited training data and an effective database expansion method. For the detection and recognition, Mask R-CNN [14] was utilised; thus, the recognition method is not new, but the focus was only on improving the training procedure.

In [15] three widely used single-stage detectors, namely YOLOv3 [16], RetinaNet [17], and SSD [18], were compared on a custom pills dataset, resulting only in minor (circa 2%) differences, but all above 0.80, in mAP. As these detectors are constantly and rapidly evolving, we also use the YOLOv8 model (in Section 3.3), but only for segmentation purposes. The reason for such detectors' limited usage is an important limitation of generic classification methods when used in few-shot problems. When an instance of a new class arrives, which has a non-negligible probability in the case of pill dispensing applications, knowing that the possible number of pill classes is over 10,000 in a typical country, the new classes should also be trained and recognised, preferably without costly efforts. Moreover, retraining deep neural networks (DNNs) can lead to catastrophic forgetting [19], although there are techniques to eliminate this effect, such as discussed in papers [20] and [21]. Furthermore, the challenges posed by the unbalanced class distribution when dealing with a low number of instances of new-coming classes cannot be overlooked. This issue, coupled with the time and/or GPU facilities needed for the retraining process, underscores the complexity of the problem.

As an alternative to generic classification approaches, metric learning approaches create an embedding of input information that can easily be used to compare queries and candidates.

If the metric model is trained on a large enough dataset representing well the possible features of possible objects, even newly arrived class instances can be classified without new training procedures, using comparisons with features of reference images.

The winner model of the aforementioned NLM challenge [4] was a metric model and was further improved in [6, 7]. This enhanced model applied a two-phase training approach: in the first phase, separate data streams, responsible for the embedding of RGB, texture, and contour information, were trained with triplets (inputs of anchor, positive, and negative samples), then in the second phase the output vectors of the streams were fused with a unique OCR stream. The OCR stream was responsible for text localisation, geometric normalisation, and feature embedding with the Deep TextSpotter [22]. This model showed better accuracy tested on the NIH (National Institute of Health) and the CURE datasets [6] (the description of datasets is in Section 3) than any previous model.

Our basic model uses a strategy similar to [6] but with significant modifications and improvements. First, we replaced the separate stream's OCR method with an LBP (local binary pattern) [23] descriptor stream, making a unified architecture that can be trained more straightforwardly without any language or symbol dependency. We have chosen LBP because local intensity changes can describe and distinguish characters or symbols. LBP is often applied in different vision tasks, including handwritten or printed OCR [24, 25]. We also added attention mechanisms into our models; details are available in Section 3.4. Moreover, besides using a different pill detection model, we use more enhanced backbones in our proposed solutions. Some of the consequences of these modifications are already investigated in [9], while some are given in Section 3. Besides these improvements, the main difference in our approach is the inclusion of different high-level (mostly explicit) cues, which were completely missing from the articles mentioned above.

Compared to general optical object recognition tasks, some things could be explicitly considered when dealing with medical pills:

1. Each pill is strictly designed to have a different appearance.
2. The distinguishing features can be described by high-level textual or visual features.

A straightforward way to utilise semantic distinguishing features is to extract imprinted characters. In [26], the authors trained not only RGB images of the pills but also imprinted characters for a separate neural network. In the pill recognition step, the different modules separately extract both the features of pills and their imprints while correcting the recognised imprint to fit the actual data of other features. A trained language model was applied for imprint correction. It was shown through an ablation study that the language model could significantly improve the system's ability to identify pills. The drawback of this solution is that a specific language model (including an OCR module) is required for the application, which might be different from country to country.

In addition to the above-listed special factors, [27] proposed a novel method, which is based on a hypothesis that all the medicines, in an actual capture of a set of pills, are prescribed to cure or alleviate some diseases or symptoms. Thus, there must be implicit relations between pills and diagnoses, which can be exploited during the recognition of the tablets. The proposed solution used a prescription-based knowledge graph, representing the relationship between pills. A graph embedding network extracted the relational features of pills and a framework was applied to fuse the graph-based relational information with the image-based visual features for the final classification. While we can generally accept the above-described assumption, there are several conditions which could weaken this approach: the images should contain several pills at a time; a large number of prescriptions are needed to train the graph model; adding new pills to the task requires a (long) process of collecting prescriptions.

In [28], the multi-pill detection problem in real-world settings is tackled by constructing heterogeneous *a priori* graphs incorporating three forms of inter-pill relationships, including co-occurrence likelihood, relative size, and visual semantic correlation. The idea is that first easy pills (those that do not have a remarkable resemblance to other pills) are differentiated based on their visual appearance. The visual features of those easy samples will then serve as context vectors to assist with making decisions regarding the hard ones (those which appear similar under the given conditions). While this approach handles the actual environmental effects and tablet-tablet size relations well, other features (e.g. colour, prints, score) are semantically not considered. Additionally, a large amount of prescription information (for model training) is required as in the previous [27] method.

A recent work, [29], introduces a pill recognition system designed explicitly for few-shot cases. It uses a virtual class synthesis strategy to enhance discriminative feature learning, where virtual classes serve as placeholders in the feature space for future class updates. To be backward-compatible with old classes, it applies a strategy to synthesise reliable pseudo-features of old classes. The main drawback of this mechanism is its complexity: it should take care of generating virtual classes, pseudo-feature synthesis, memory banks, knowledge distillation, and three types of loss functions - all with parameter settings.

In contrast to all these approaches, our proposed solution utilises all possible semantic information about individual pills; it avoids the use of specific models for the imprinted text or for prescription data, while it uses only generic models (to process the information leaflet of pills) and can handle few-shot classification in a self-explanatory way.

Based on the above observations, we believe that metric learning has essential advantages over other approaches. In [9], we have shown that OCR can be omitted from the streams, and in [30], we have first shown that pills' textual information can be utilised in visual metric learning. Now, we are to extend our research with the following:

- Since in some datasets, textual information is not available for word-based semantic information, we extend the model with visual features to inject semantics into the training of the metric model. We describe the methods for generating those visual descriptors (colour, embossed, engraved or painted marks and texts, score, and shape), including their custom embedding network.
- By running numerous experiments and ablation studies, we analyse the performance of the baseline and injection models (both textual and visual).
- Enhanced pre-processing is done with the pixel-wise segmentation of pills and substituting backgrounds with constant colour in the training process.
- Beside our own (publicly available) dataset, we include the CURE dataset for the experiments and compare results to best known existing method [6, 7].

Results achieved with our improved model are now ahead of our previous attempts [9, 30] and others' [6, 7].

3 | MATERIAL AND METHODS

3.1 | Overview of the proposed approach

In this chapter, we introduce our model, referred to in our article as Multi-Stream Fusion Network (MSFN) and Dynamic Margin Multi-Stream Fusion Network (DM-MSFN). The model, aimed to estimate the distance among pills, follows the multi-stream design and includes multi-head attention layers. Our main innovation lies in generating and integrating high-level textual and visual cues into metric learning to improve pill recognition. This is achieved by altering the margin of the triplet loss function, enabling a more granular metric learning process, surpassing the capabilities of traditional triplet loss models. The overview of the proposed MSFN structure is given in the upper part of Figure 1; the bottom part illustrates the training mechanism, including the text and image processing parts. In Algorithm 1, subsequent training steps are described.

In the next section, we describe the datasets we utilised during our research.

3.2 | The OGYEIV2 and CURE datasets

Several image datasets, such as those in [4, 6, 31], are available for pill recognition. In our experiments we have chosen two datasets: one is the CURE image database [6] and the other is our custom dataset, entitled OGYEIV2 [30, 32].

The purpose behind choosing the CURE dataset is twofold: it is well-documented and accessible and has been widely used in research papers. The dataset contains images of single pills with different backgrounds and varying lighting conditions. We have chosen this dataset for comparison since results of one of the best metric-based methods [6] are available for it. CURE is divided into reference and consumer quality (query) images. It is important to note that images are also separated by their side (top or bottom of a pill). Reference images are syntheti-

ALGORITHM 1 Training of DM-MSFN.

Input: Dataset X , Triplet generator $\tau_g(\cdot)$, Epochs E , Feature streams $f_{rgb}(\cdot)$, $f_{lbp}(\cdot)$, $f_{contour}(\cdot)$, $f_{texture}(\cdot) \subseteq \mathcal{S}$

Output: Trained MSFN $f_{msfn}(\cdot)$

Image preprocessing (Section 3.3):

- 1: **for** batch X_b in X **do**
- 2: pill detection, segmentation, background replacement: $X'_b = \phi(X_b)$
- 3: **end for**

Extracting high-level textual cues for each class (Section 3.5.1):

- 1: Extract sentences from leaflets.
- 2: Create word embedding with `hu_core_news_lg`.

Extracting high-level visual cues for each class (Section 3.5.2):

- 1: Encode visual features of reference images.
- 2: Train fully-connected network to generate visual embeddings.

Phase 1: Feature streams training (Section 3.4)

- 1: Initialise feature streams
- 2: Initialise set for hard triplets: $H_{str} \leftarrow \emptyset$.
- 3: **for** epoch e in E **do**
- 4: **for** batch X'_b **do**
- 5: Generate triplets $T_a \leftarrow \tau_g(X'_b)$
- 6: **for** each triplet (I_a, I_p, I_n) in T_a **do**
- 7: Train Feature streams $\mathcal{S}$ independently with triplet loss $\mathcal{L}_{tri}(\theta)$
- 8: **end for**
- 9: **end for**
- 10: **end for**

Hard triplet mining (Section 3.4):

- 1: **for** each triplet (I_a, I_p, I_n) in T_a **do**
- 2: **if** $D(f_\theta(I_a), f_\theta(I_n)) - D(f_\theta(I_a), f_\theta(I_p)) < m$ **then**
- 3: $H_{str} \leftarrow H_{str} \cup \{(I_a, I_p, I_n)\}$
- 4: **end if**
- 5: **end for**

Phase 2: Fusion of streams (Section: 3.4.2)

- 1: Freeze weights of all feature streams $\mathcal{S}$
- 2: Initialise fusion network $f_{fusion}(\cdot)$
- 3: **for** epoch e in E **do**
- 4: **for** batch X_b in H_{str} **do**
- 5: Train $f_{fusion}(\cdot)$ with triplet loss $\mathcal{L}_{tri}(\theta)$
- 6: **end for**
- 7: **end for**

cally derived and generated from consumer images by replacing the backgrounds with homogeneous colours. We have found some minor flaws in CURE: some pills are listed twice or more in different classes (with different photographs). CURE lacks pixel-wise segmentation annotations and textual descriptions of pills.

Due to the above-mentioned limitations of the CURE dataset we also evaluate some models on our own pill dataset, called OGYEIV2.

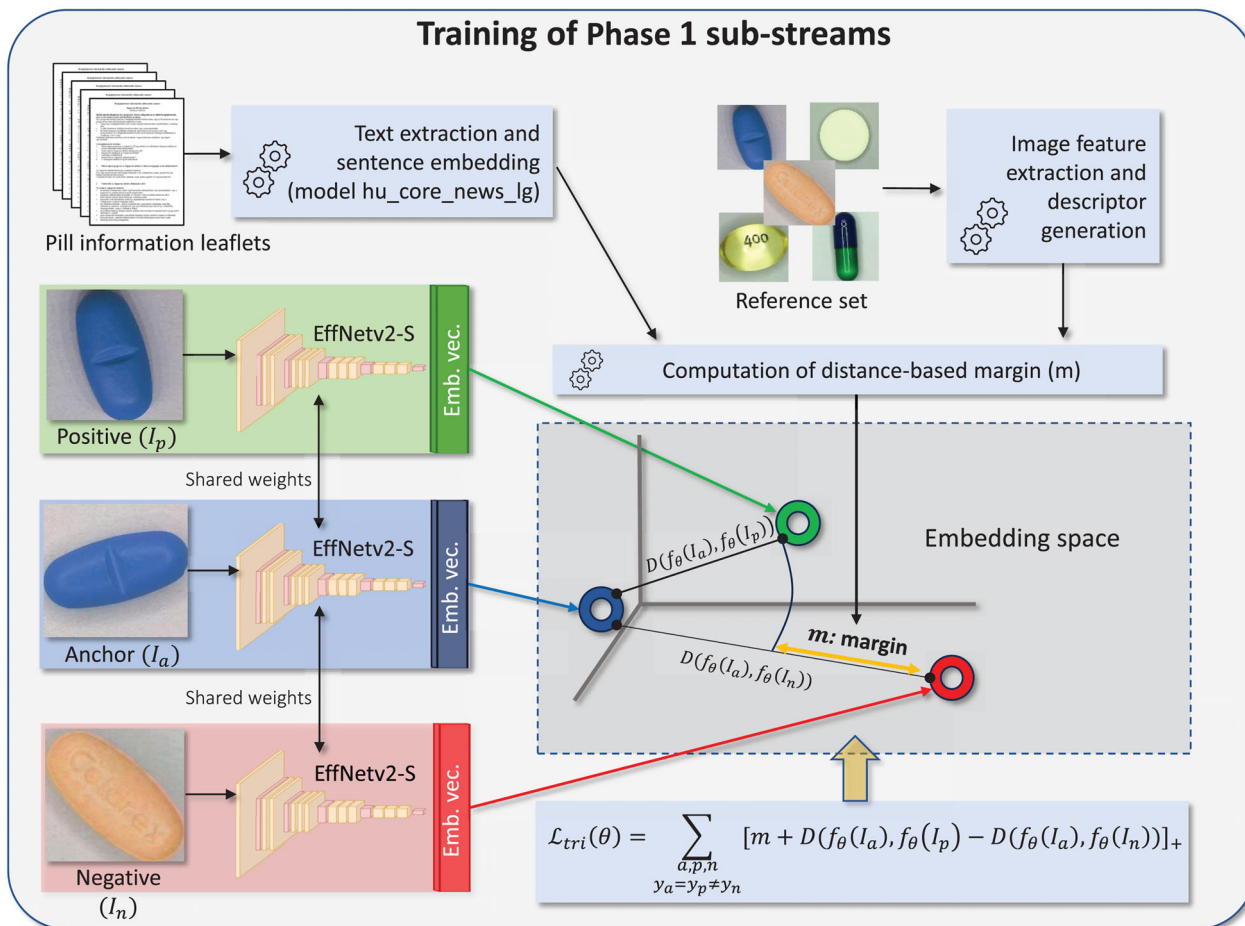
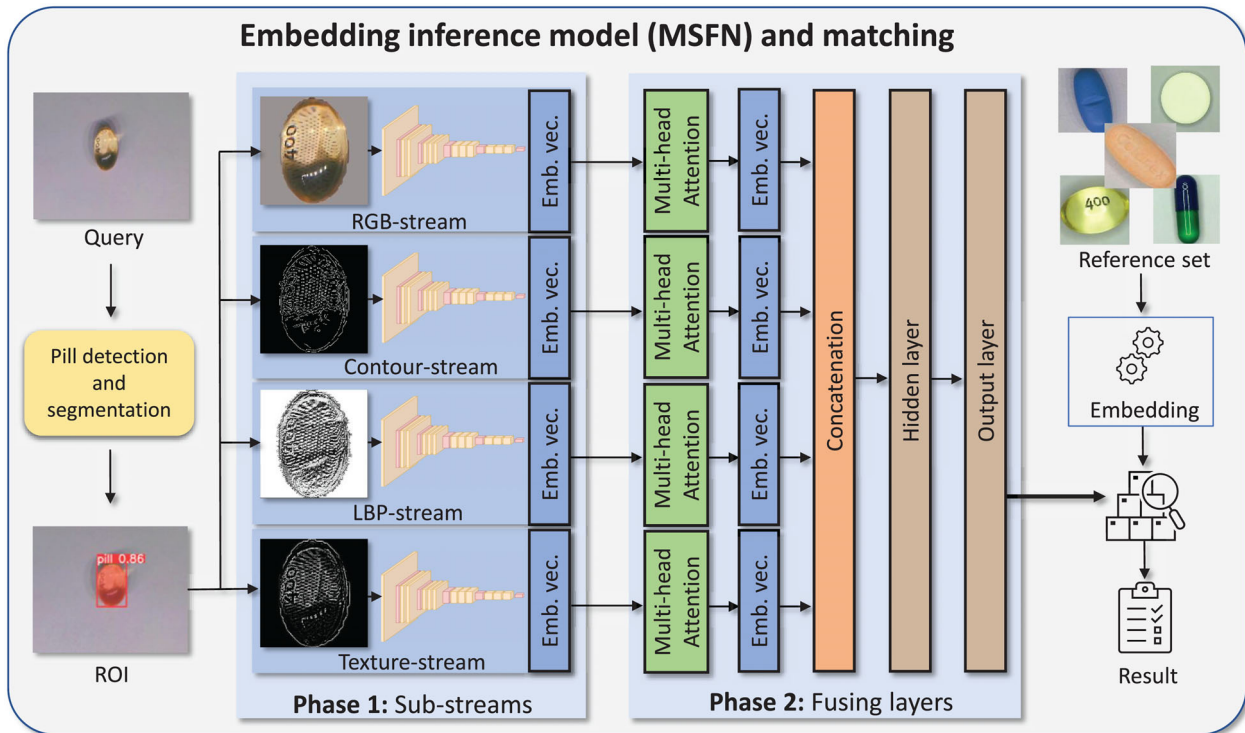


FIGURE 1 Overview of the proposed Multi-Stream Fusion Network. Upper part: Data flow during inference. Bottom part: Illustration of the training mechanism on sub-streams; the same approach is applied on the fusing layers. The explanation of distance function D , the value of m , and triplet loss $\mathcal{L}_{tri}(\theta)$ is in Section 3.6.



FIGURE 2 Images from the two datasets employed in the study. The first row shows image pairs from the CURE dataset, with the first images in each pair being a reference-quality image and the second image being a consumer-grade image. The second row presents image pairs from the OGYEIV2 dataset, where the first image in each pair was captured using an upper lighting, while the second image was captured using a side lighting. The unwanted effects of the different lighting are clearly observable.

TABLE 1 Comparison of the main properties of the CURE and OGYEIV2 datasets.

	CURE	OGYEIV2
Number of pill classes	196	112
Number of images	8973	4480
Image resolution	800×800 2448×2448	3246×2019 (undistorted)
Instance per class	40-50	40
Backgrounds	6	1
Segmentation labels	No	Yes
Free-text description	No	Yes

The creation of OGYEIV2 can be described the following way:

- The pill images were captured by an ELP 8.3 MPx USB camera (Sony IMX415 sensor up to 3840 × 2160 pixels of native 4K resolution) using an 8 mm, F1.8 optic. We applied camera calibration to eliminate optical distortions; after undistortion, the image resolution became 3246 × 2019 pixels.
- The same homogeneous grey background was used for each drug picture.
- Half of the images were taken under an upper LED ring lamp with a diffuser attached. In this configuration, pills gave a pleasant overall appearance without noticeable shadows (but some engravings almost disappeared; see the second row of Figure 2).
- Another half of the images were recorded using a side-mounted LED strip light, making the engraved surface patterns sharply visible.

While images in OGYEIV2 have similar backgrounds (easily achievable in daily conditions), the two different light settings make sign and score-based recognition difficult (similar to real-life consumer applications). The main characteristics of the two datasets are shown in Table 1.

TABLE 2 Pill segmentation results on the 20% of CURE consumer images.

	U-Net (35M)	W ² -Net (2M)	YOLOv8m-seg (27.3M)
IoU	0.90	0.94	0.956

According to our knowledge, OGYEIV2 is the only publicly available pill dataset that contains not only images but also free-text visual appearance descriptions taken from the original leaflets of the medicines. These leaflets, besides others, contain a general description of the visual appearance of the pills. To show a short example of the textual descriptions, we give the original sentence (in Hungarian) from the leaflet of the medicine named Advil Ultra Forte and its English translation:

- Hungarian: *Színtelen, tiszta folyadékot tartalmazó, átlátszó, ovális lágy kapszulák fekete "400" felirattal.*
- The English translation: *Colourless, containing clear liquid, transparent, oval soft capsules, imprinted with black "400" lettering.*

Since the leaflet content is strictly structured, we used NLP to gather the relevant morphological information (2 or 3 sentences per leaflet).

3.3 | Preprocessing: Detection and segmentation of pills

Three errors can occur when detecting and recognising multiple or single pills from an image. False detection of other objects, the pill is found but misclassified, or even worse, the pill is not detected. When single-stage detectors (such as YOLO or SSD) are considered, the whole training process (detection, localisation, recognition) is carried out in a single step. However, metric learning only deals with the recognition of already detected objects. That is why the pills are detected and localised in a pre-processing step. The model has to contend with various factors: noisy backgrounds, challenging lighting, perspective distortions, and scaling from a distance. These problems greatly affect the accuracy of the segmentation results, thereby, the overall process of pill recognition [6]. Pill backgrounds provide very little or no useful information and may even degrade the quality of the training process, so they are considered noise. Although it is crucial to separate the pill regions from the background, which allows the tablet recognition network to be trained and infer from tablet images without the background details, the primary aim of our article is not the segmentation itself. Similarly to [6, 26, 27], we focus only on the recognition of already segmented pill images and the background is replaced with a constant grey value like in [6]. Unfortunately, for the segmentation of pills we could not reproduce the W²-net architecture, described in the supplementary material of [6], and neither it is available online. Thus, we have chosen the YOLOv8 [33] network, which has comparable instance segmentation capabilities (see our results in Table 2).

For the training of YOLOv8, we used reference images, while for the testing, 20% of consumer images were selected from

CURE, similarly to [6]. Both in the augmentation and the testing process, images were pixel-wise annotated. The following augmentation operations were performed:

- white balance changes to slightly alter the colour of the images,
- Gaussian smoothing to simulate out-of-focus,
- brightness adjustments: turning the image either brighter or darker,
- geometric transformations: rotation, shifting, and scaling,
- changing the background of pills using the Describable Textures Dataset (DTD) [34].

These steps are similar to the augmentation applied in [6]. During the generation and testing of the OGYEIV2 dataset, we found that ambient light colour can significantly affect recognition performance. Thus, white balance changes were introduced additionally.

The final number of images after the expansion was 105,661, which were further divided into training and validation sets, while testing was performed on the 20% of the consumer images, as mentioned above. In [6], authors reported 0.94 as the highest segmentation IoU value of their W^2 -network, while we could achieve a slightly higher value of 0.956 (see Table 2, which also includes the results with vanilla U-Net [35]). Our metric learning models use the pill images detected by this trained YOLOv8 model.

3.4 | Multi-stream metric learning

To guide the DNNs in finding the appropriate conceptually designed visual features, we applied a multi-stream network architecture comprising four sub-streams of very similar structures. Each sub-stream was subsequently integrated with multi-head attention modules [36], followed by concatenation and fusion to generate the final output vector. To reach the best performance, we trained our network in two consecutive phases: first, the sub-streams, then the fusing block, including the attention layers. The architecture of MSFN is shown in the upper part of Figure 1.

3.4.1 | Feature streams

The idea behind using various sub-streams is to encourage the extraction and utilisation of diverse image features that may be advantageous in different situations and for different tablet classes. Our approach implements the following sub-streams:

1. Contour-stream: beginning with a grayscale segmented image I_{grey} , a contour image I_{con} is generated through a series of steps. First, Gaussian smoothing with a kernel size of 7×7 is applied to I_{grey} . Subsequently, the Canny edge detector is employed.
2. LBP-stream: to avoid the usage of language-specific streams (like the OCR stream in [6]) and to have a homogeneous sub-stream architecture, we use LBP [23] descriptors. LBP,

known for its effectiveness in computer vision tasks like OCR, was computed from grayscale images I_{grey} to yield the LBP image I_{lbp} .

3. RGB-stream: as for RGB images I_{rgb} , colour input images were simply passed to the embedding deep neural network.
4. Texture-stream: texture images I_{tex} were generated by initially applying a Gaussian filter to the input grayscale image I_{grey} , followed by subtraction of the resulting image I_{res} from the original one.

Each sub-stream received a segmented image as an input. The output of the data streams consisted of 128 neurons, except for the RGB stream, which comprised 256 neurons.

To train the stream networks, Siamese neural networks (SNNs) were utilised with three inputs, namely anchor, positive, and negative images (denoted by I_a , I_p , and I_n , respectively). In contrast to [6], where the authors created a lightweight CNN network with only 9 million parameters, we opted for the EfficientNetV2-S (small-sized) network [37], exploiting its optimised architecture. Figure 1 bottom part shows the training mechanism applied for sub-streams.

All four sub-streams were trained independently for metric embedding, using a triplet mining strategy known as batch-all. This strategy considers all possible sample triplets within the current mini-batch during the training process.

During the initial phase, samples deemed challenging (called hard samples) were collected and preserved for later use in the second phase (when training the fusing layers)—this is called the batch-hard strategy. These hard samples are categorised into two groups: hard positive and hard negative. Hard positive patterns belong to the same class but exhibit feature representations likely to be tightly located in feature space, making it challenging for the model to discriminate effectively. Conversely, hard negative samples belong to different classes yet are close to each other in the feature space, which poses a challenge as the model may misclassify them as belonging to the same class.

3.4.2 | Fusion of streams

In the second training phase, we implemented and integrated multi-head attention modules [36] after each stream. To aggregate the information from these modules, the resulting output vectors were concatenated. Subsequently, a hidden layer and an output layer were combined to generate the final embedding, resulting in a 640-dimensional vector. It is worth mentioning that at this stage, the previously trained stream networks were frozen in order to preserve their existing weight parameters. The fusion network was trained solely with hard samples with the same approach as the individual streams (bottom part of Figure 1).

3.5 | Adding high-level cues

High-level cues describe features related to high-level semantic meaning that humans can easily describe and understand. Since pills are designed with the intention that people can distinguish

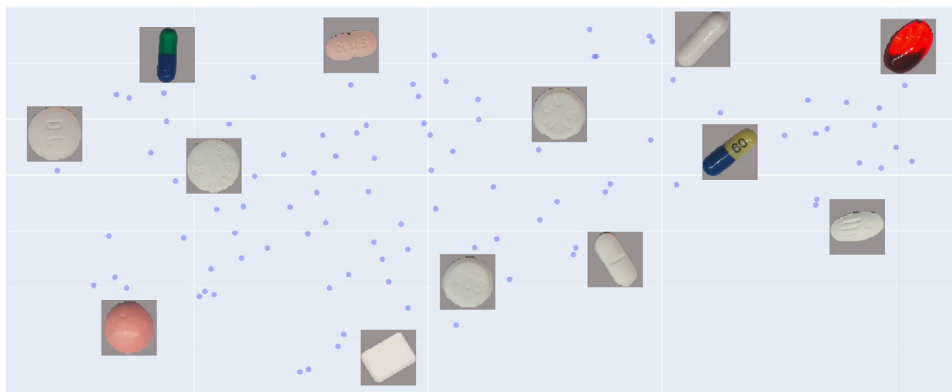


FIGURE 3 Visualisation of pill samples, from the OGYEIV2 dataset, in 2D metric space based on word embeddings using high-level textual cues.

them, we are to explicitly use this information to enhance the baseline MSFN metric model. The following semantic features were utilised as high-level visual and textual cues: colour, embossed, engraved or painted marks and texts, score, shape, and free text descriptions from leaflets.

3.5.1 | High-level textual cues based on word embedding

To obtain textual semantic information, we used information from the leaflets of the pills as described in Section 3.2. Extracted sentences were tokenized, paying close attention to high-level features such as pharmaceutical form, colour, shape, edge, imprint, embossing or engraving. In this study, to generate high-level semantic embedding of pills, we used the `hu_core_news_lg` network, which is a CNN-based, powerful and pre-trained language model that can be found in the Python package called HuSpaCy [38]. The `hu_core_news_lg` model is trained on the broad Hungarian Webcorpus 2.0, comprising over 9 billion words from newspapers. Integrated into spaCy's language model pipeline, it provides tokenization, part-of-speech tagging, lemmatization, and dependency parsing. The designation “lg” in its name signifies that it is a large model with 200,000 distinct word vectors and 300-dimensional feature vectors (called florets). To be able to plot the word embeddings, we reduced their dimensionality to 2D by employing t-Distributed Stochastic Neighbor Embedding (t-SNE) [39], results can be seen in Figure 3.

3.5.2 | Extracting and embedding high-level visual cues

In those cases when no textual information is available, we can still produce high-level visual cues from good-quality pill images (such as reference images of the CURE dataset) or any labelling information that might be available for the classes. Now, we describe the steps to generate the visual descriptors in the following order: colour, markings, score, and shape.

TABLE 3 Encoding pill surface signs.

	$S(1)$	$S(2)$	$S(3)$
Embossed/engraved pill	0	$\frac{1}{2}$	1
Pill with printed text	$\frac{1}{2}$	0	1
Plain pill	1	1	0

To generate the colour feature vectors, we selected two 30×30 pixel regions on each pill, ensuring that they were favourably spaced apart (for the sake of capsules often with two different colours), and sampled colours from these regions. Subsequently, the RGB values obtained from these samples were averaged and converted to the CIELAB colour space [40]. This perceptual colour space is used because of its relevance to colour-related visual tasks. This process yielded a 12-dimensional vector for the OGYEI and CURE two-sided datasets and a 6-dimensional vector for the CURE one-sided dataset. While using the two colour values for pills for comparisons is not straightforward, we expect the subsequent embedding network to handle complex cases to an acceptable extent.

The markings on the tablets can be categorised into one of the three classes: embossed or engraved, imprinted, and empty. We encoded these classes into $S(i) \in \mathbb{R}^3$ with the scheme shown in Table 3. The idea is to have a slightly more sophisticated and expressive coding pattern than one-hot encoding, used as the input of the subsequent embedding network.

This process resulted in 6-dimensional vectors for the OGYEIV2 and CURE two-sided datasets and 3-dimensional vectors for the CURE one-sided dataset.

The same approach was applied to the score of the pills. Score defines how many parts a tablet can be broken into (i.e. non-breakable, halvable, quarterable). The corresponding encoding scheme for the score (breakability) descriptor vector $B(i) \in \mathbb{R}^3$ is given in Table 4.

Shape descriptors for the pills, to approximate and characterise the distinctive shapes of the pills, were generated using elliptic Fourier descriptors (EFD) [41]. This step required pre-processing of the images, which involved Gaussian blurring and



FIGURE 4 Illustration of shape types of pills from the CURE dataset.

TABLE 4 Encoding scores (breakability).

	$B(1)$	$B(2)$	$B(3)$
Not breakable pill	0	1	1
Pill breakable to half	1	0	$\frac{1}{2}$
Pill breakable to quarter	1	$\frac{1}{2}$	0

Canny edge detection. It was followed by a morphological opening, after which the most prominent contour, based on area, was identified and used for further analysis. For each pill image, the EFD was calculated with a degree of approximation set to 10, resulting in a vector of 10×4 numbers:

- a_n : Amplitude of the cosine component in the x-direction for the n_{th} harmonic.
- b_n : Amplitude of the sine component in the x-direction for the n_{th} harmonic.
- c_n : Amplitude of the cosine component in the y-direction for the n_{th} harmonic.
- d_n : Amplitude of the sine component in the y-direction for the n_{th} harmonic.

We applied normalisation to achieve rotation and size invariance, which also means that the first three values of the coefficients could be omitted from further calculations (since normalised values were equal).

To validate the accuracy of the generated Fourier descriptors, we manually categorised pills into eight groups (as in Figure 4). We calculated the average of the Fourier descriptors within each, and finally, we computed the Euclidean distance between these groups as illustrated in Figure 5.

The capsule's shape seems similar to an oval and a doubleround. The corresponding numbers in Figure 5 are 0.14 and 0.16, respectively. On the other hand, a capsule differs from the round, pentagon, and triangle shapes, where the distances are 0.52, 0.53, and 0.56. Take another example: while the distance between a pentagon and round shapes is 0.12, the distance between a pentagon and a doubleround is 0.41, which is close to the intuitive judgement of the observer.

In order to combine the descriptors of the above properties in a compact form, we designed and implemented a fully connected embedding network with a descending architecture of 4 layers, where the number of neurons was 100, 80, 60, and 10 in succession. The input to the network consisted of the concatenated descriptive vectors defined above, while contrastive loss [42] was chosen to achieve metric embedding. The train-

ing ran for ten epochs, the batch size was 16, and the Adam optimiser was chosen with a learning rate of 5×10^{-4} and a weight decay of 1×10^{-3} . To visualise the embeddings of pills of the OGYEIV2 dataset, the 2D representation, generated with t-SNE, see Figure 6.

The embedding vectors, based either on free text, labels, or image data, will be used in the training of the multi-stream metric model. Embedding information will be used in the loss function discussed in the next section.

3.6 | A triplet loss function for injecting high-level cues

We have already seen the illustration of the sub-stream training in the bottom part of Figure 1. We use three inputs to learn the metric space representation of images. Each triplet has an anchor sample (I_a), a sample from the same class as the anchor (positive sample, I_p), and one from a different class (negative sample, I_n). The anchor and the positive sample must originate from the same class but should not be identical images. The goal is to learn the mapping function (from the image domain to the E dimensional metric space):

$$f_{\theta}(I) : \mathbb{I} \rightarrow \mathbb{R}^E. \quad (1)$$

by optimising the model parameters (θ) to minimise the distance between the feature space representation of the anchor and the positive sample (I_a, I_p), and maximising the distance between the feature space representation of the anchor and the negative sample (I_a, I_n) simultaneously. This can be achieved by the following triplet loss function:

$$\mathcal{L}_{tri}(\theta) = \sum_{\substack{a,p,n \\ y_a=y_p \neq y_n}} [m + D(f_{\theta}(I_a), f_{\theta}(I_p)) - D(f_{\theta}(I_a), f_{\theta}(I_n))]_+, \quad (2)$$

where m represents the margin, which ensures a certain separation between positive and negative samples, and y_i denotes the pill class label for the i_{th} sample. The function $D(f_{\theta}(I_i), f_{\theta}(I_j)) : \mathbb{R}^E \times \mathbb{R}^E \rightarrow \mathbb{R}$ indicates the metric function that calculates the Euclidean distances between the two images I_i and I_j in the embedding space. In triplet metric learning models, m is chosen to be constant [6, 9]; we proposed to dynamically set its value in our recent paper [30]. Our proposal is more suitable and intuitive than the so called Log-ratio loss [43], where the margin also changes dynamically but does not assume positive and negative samples, which is a rather strict category in the case

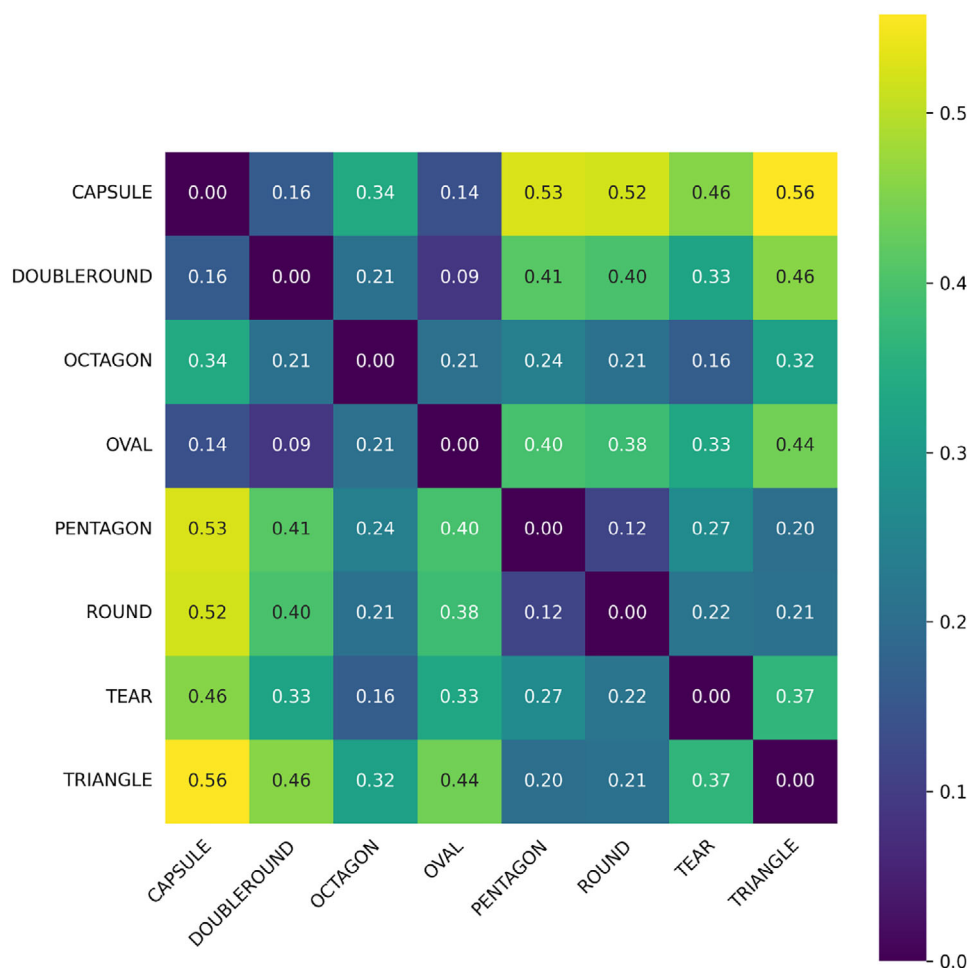


FIGURE 5 Pairwise Euclidean distances between shape groups of pills of the CURE dataset based on contour elliptic Fourier descriptors.

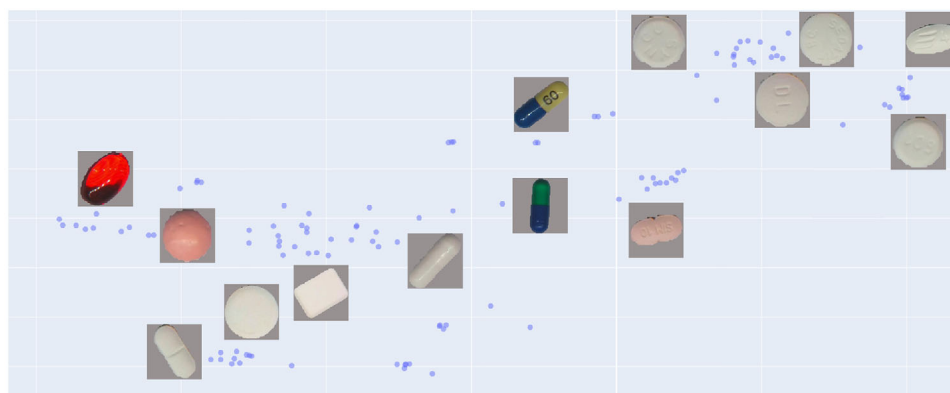


FIGURE 6 Visualisation of pill samples, from the OGYEIV2 dataset, in 2D metric space based on high-level visual cues.

of pill recognition. The proposed dynamic margin triplet loss function (DMTL) uses a margin continuously set according to the conceptual difference of the actual input samples. While in our recent article [30], the value of m was based on the distances of the word embeddings, now we are extending the concept to visual data as described in Section 3.5.2. The idea is that since we

have high-level cues about the visual appearance of classes, we can use this information to position the objects in the feature space more gradually if we set m higher to less similar objects and lower for more similar ones:

$$m(a, n) = \alpha \cdot d_i^{Norm}, \quad (3)$$

where $\alpha = 0.2$ and the normalised distance d_i^{Norm} is determined by the distance between the two samples a and n based on the high-level cue embeddings:

$$d_i^{Norm} = 1 + \frac{(u-1) \cdot (d_i - d_{min})}{d_{max} - d_{min}}. \quad (4)$$

Here d_i represents the Euclidean distance of samples, d_{max} stands for the maximum distance, while d_{min} denotes the minimum distance within a row of the distance matrix (formed by distance values of classes), with u as the upper limit.

4 | EXPERIMENTS AND RESULTS

It is important to note that in this research, we conduct one- and two-sided tests for the CURE dataset, while for OGYEIV2, the tests are only two-sided. (The two-sided test considers both sides of a pill as belonging to the same category, whereas the one-sided test treats the two sides as belonging to two separate classes, doubling the number of classes).

To increase the statistical robustness of the evaluation, we employed 5-fold cross-validation. Throughout the assessment process, we followed the conventional methodology used for metric learning models: the reference and consumer images underwent the metric embedding procedure, then the generated vectors were compared to each other using the Euclidean distance (see the top-right part of Figure 1) to find the best matching. For all 5 folds, Top-1 and Top-5 accuracies were calculated, averaged, and then converted to percentages.

4.1 | Use-cases

Two use-cases were examined considering the selection of the reference set:

1. Traditionally, the reference set includes all drug classes in the dataset. We refer to this as the *Complete Reference Set*, also referred to as *gallery set* in other publications. In this case, we selected one fold from the consumer classes for queries during the inference process. However, all the reference classes are present in the candidate set (thus, the model has the chance to miss-classify a query with any of the classes - trained or not trained).
2. In the second type of reference set, we include only such classes that are present in the actual query fold, ensuring that the model compares identical consumer and reference classes. Naturally, this approach performs better, reflected in both Top-1 and Top-5 accuracy. We refer to this smaller reference set as *Actual Fold Reference Set*, and it resembles use-cases when, during pill recognition only those classes are considered as candidates which are also in the query set (naturally with different sample images).

4.2 | Hyperparameter settings

Training the data streams also involved finding the best possible hyperparameter settings. Nowadays, it has become common practice to use Ray Tune [44]—or any other hyperparameter tuning library—to find the proper parameters for the given task. However, implementing this in our case would have been way too cumbersome, and tuning the networks for the best parameters would have consumed too much time. Therefore, the hyperparameters were determined empirically, and the results were still acceptable.

After the segmentation, all images were resized to 224×224 , and the pixel values were normalised to the 0–1 range. For our OGYEIV2 and the CURE datasets, the parameters were almost the same: in the case of the CURE dataset, we employed the Adam optimiser, the learning rate was set to 1×10^{-3} for all four streams; the batch size was 32; and the margin of the triplet loss function was chosen as 0.5. Learning rate scheduling was applied at every five epochs, where gamma was chosen to be $1/3$. The weight decay was 1×10^{-5} . In the OGYEIV2 dataset, only the learning rate was set differently to 1×10^{-4} . As for the dynamic margin triplet loss (DMTL) model, we applied a weight decay regularisation with a coefficient of 1×10^{-4} and changed the learning rate to 1×10^{-4} for both datasets - everything else remained unchanged. In all cases, each model was trained for a total of 25 epochs, and only the best weight file was saved.

For the fusion phase, we also tried to apply the best possible hyper-parameters: batch size was kept at 32; the learning rate was changed to 2×10^{-4} for better stability; the weight decay was initialised to 1×10^{-8} . In this phase, we again implemented the learning rate scheduling, which involved adjusting the initial learning rate at every two epochs using a gamma value of 0.1. The network was trained over ten epochs. As with the stream-phase, only the best weight file was saved.

We substituted the final linear classifier layer of the EfficientNetV2 models with a customised linear layer tailored to our task specification. After this adjustment, the number of parameters in the EfficientNetV2-S model comprised approximately 20.4 million for the contour, LBP, and texture streams and about 20.5 million for the RGB stream. Thus, the entire Fusion network encompassed 82.5 million parameters. We used an Nvidia Quadro RTX 5000 and an Nvidia RTX A6000 GPU card to train and test the models.

4.3 | Multi-stream processing without high-level cues

In this series of tests, the traditional triplet loss function was applied with static margin ($\alpha = 0.2$ as found best in [6]). We compare our MSFN model against the one published in [6, 7], trained on the CURE dataset, in Table 5. As for the OGYEIV2 dataset, we can compare the performance to the measurements of [30], data are in Table 6. More details of these comparisons are in Section 5.

TABLE 5 Average performance over 5 folds on the CURE dataset using static margin.

	CURE one-sided			CURE two-sided		
	Top-1 accuracy MS in [6]	Top-1 accuracy MSFN	Top-5 accuracy MSFN	Top-1 accuracy MS in [6]	Top-1 accuracy MSFN	Top-5 accuracy MSFN
Actual Fold Reference Set	N.A.	87.5	98.52	N.A.	88.63	98.88
Complete Reference Set	68.30	69.96	96.90	84.6	85.53	97.61

TABLE 6 Average performance over 5 folds on the OGYEIV2 dataset using static margin. First two columns contain new results, while the last two ones are from paper [30] for comparisons.

	OGYEIV2			
	Top-1 accuracy	Top-5 accuracy	Top-1 accuracy in [30]	Top-5 accuracy in [30]
Actual Fold Reference Set	99.08	100	91.72	98.125
Complete Reference Set	96.12	100	N.A.	N.A.

TABLE 7 Average performance over 5 folds on the OGYEIV2 dataset using dynamic margin (DM-MSFN model) with textual cues. First two columns contain new results, while the last two ones are from paper [30] for comparisons.

	OGYEIV2			
	Top-1 accuracy	Top-5 accuracy	Top-1 accuracy in [30]	Top-5 accuracy in [30]
Actual Fold Reference Set	99.23	100	93.56	99.37
Complete Reference Set	96.85	100	N.A.	N.A.

4.4 | Multi-stream processing with textual cues

In this test, we applied the DMTL with textual cues (defined in Section 3.5.1). As mentioned above, this test could only be performed on the OGYEIV2 dataset (as no textual information was included in the CURE database). Average results of the five folds are in Table 7, including the previous results of [30].

4.5 | Multi-stream processing with visual cues

In this final test series, DMTL was again applied, this time with visual cues (as defined in Section 3.5.2). All three datasets were utilised (e.g. one- and two-sided tests for CURE and two-sided tests for OGYEIV2). Results of all these tests are in Table 8.

4.6 | Ablation studies

In-depth comparisons were made with ablative models for both the static and dynamic margin models. The performance of individual streams was compared, and the results are summarised in Table 9.

The results clearly show that the RGB stream is the most dominant because it contains all raw details and has the highest number of output nodes of all streams. However, this RGB stream alone is not sufficient, and as the results show, when fused with the other three streams, the results are continuously improved. This improvement of the static model is 5.78% for OGYEIV2, 2.15% for CURE one-sided, and 6.28% for CURE two-sided tests.

We also ran tests where we omitted specific streams and trained the network accordingly. Results, only given for the static margin variant and on the OGYEIV2 and the CURE one-sided datasets, are summarised in Table 10. Both tables support our decision to include the LBP stream: after the RGB stream, LBP showed the highest accuracy (see Table 9), and its removal caused the highest loss (see Table 10).

5 | DISCUSSION

Under one-sided settings, the Top-1 accuracy reported by [6] was 68.3% for the publicly available CURE dataset. The proposed baseline (static) MSFN model has outperformed this accuracy by 1.66%. In the two-sided settings, the numerical advantage of the baseline MSFN model is only 0.93%, which is somewhat reasonable, since all numbers are higher (and closer to 100%) in two-sided settings. It is also worth noting that the proposed MSFN model is uniform and does not require a language-specific OCR sub-stream, which had a significant positive contribution to the accuracy of the model of [6].

Using high-level visual cues (DM-MSFN), we could further improve the accuracy with 1.23% and 0.67%, respectively, on the CURE dataset. Thus, the overall improvement is 2.89% and 1.6%.

During the assessments on the OGYEIV2 dataset, we could compare the contribution of the textual and visual high-level cues. Considering the Top-1 accuracy value of the static model with 96.12% (a high value itself), the improvement is 0.71% with the visual and 0.73% with purely textual cues. The 99.23% Top-1 accuracy of the DM-MSFN model on the Actual Fold

TABLE 8 Average performance over 5 folds on the datasets using dynamic margin (DM-MSFN model) with visual cues.

	OGYElv2		CURE one-sided		CURE two-sided	
	Top-1 accuracy	Top-5 accuracy	Top-1 accuracy	Top-5 accuracy	Top-1 accuracy	Top-5 accuracy
Actual Fold Reference Set	99.14	100	89.02	98.67	89.20	99.54
Complete Reference Set	96.83	100	71.19	97.28	86.20	97.95

TABLE 9 Individual stream processing with static margin.

		OGYElv2		CURE one-sided		CURE two-sided	
		Top-1 accuracy	Top-5 accuracy	Top-1 accuracy	Top-5 accuracy	Top-1 accuracy	Top-5 accuracy
Individual streams	Contour	55.83	89.42	38.04	74.39	44.67	82.48
	LBP	80.96	99.62	53.98	86.47	64.85	93.62
	RGB	90.34	99.88	66.15	95.41	79.27	96.92
	Texture	78.35	98.65	47.83	83.12	56.25	88.68

TABLE 10 Impact of domain related features with static margin.

		OGYElv2		CURE one-sided	
		Top-1 accuracy	Top-5 accuracy	Top-1 accuracy	Top-5 accuracy
Impact of domain related features	w/o contour	94.69	99.84	68.42	95.77
	w/o LBP	93.14	99.70	66.57	94.91
	w/o texture	93.89	99.89	67.83	95.51

Reference Set means that almost perfect classification is possible if images are good quality and candidates are only from those classes known during training. We also present our earlier results from [30] in Tables 6 and 7. While there is a moderate improvement in Top-5 accuracy, Top-1 accuracy shows a significant improvement. The reason for this is the pixel-wise segmentation of pills and substituting backgrounds with constant colour in the training process. Furthermore, we did not use the batch-hard strategy in [30].

While in [29] (mentioned in Section 2), the primary purpose was to make a new method for incremental learning, we can still partly compare their results to ours using the CURE dataset. Considering their setup with 131 classes, including 40 with only five shots, they obtained 84% Top-1 accuracy. In comparison, in our setup with 196 classes, including 39 with only two shots, we reached 86.2% (given in the 5th column of Table 8).

In Table 11, we divided the OGYElv2 dataset into two subsets based on the direction of the light illuminating the pills. It is important to note that these results are not fully comparable with those in Table 9, as only half of the dataset was used in each training and evaluation (images with upper light and images with side light). First, it is clear that upper light images show higher Top-1 performance compared to side light ones for all streams except for the contour stream. Two example pills well illustrate this in the second line of Figure 8: important parts

(especially scores, engravings, and embossing) are disappearing from upper light images, reasoning the performance drop of the contour stream in such cases. One would think that the contour stream could be omitted since its features are similar to the texture stream's (see again Figure 8). Unfortunately, by the comparison of Tables 5, 6, and 10, we see that leaving out the contour stream results in circa 1.5% loss in accuracy.

In Figure 7, we present some difficult examples of pill classification from the OGYElv2 dataset. The image pairs on the left illustrate classifications performed without high-level cues, whereas the image pairs on the right were produced with the DM-MSFN model. In the first two rows, only the static margin variant model (where no high-level cues were utilised) misclassified the pills, whereas the dynamic margin variant correctly classified them. In the third row, both variants failed to classify the query pill accurately. Since the bounding boxes were scaled to the same size, we expect that without size information, no visual method could always reach 100% accuracy.

We also evaluated the inference time of the individual modules within the proposed MSFN. The reported results represent the mean inference time per single image. Evaluations involved CPU (AMD Ryzen 5 5600X) and GPU (Nvidia RTX5000) devices. Results can be seen in Table 12, showing the model is applicable on a few pills, even with a CPU system, and applicable on a larger scale with GPU.

TABLE 11 Individual stream processing with static margin on the separate OGYEIV2 dataset.

Ablative models		OGYEIV2 side light		OGYEIV2 upper light	
		Top-1 accuracy	Top-5 accuracy	Top-1 accuracy	Top-5 accuracy
Individual streams	Contour	49.15	88.81	45.76	88.52
	LBP	60.04	94.66	61.74	94.85
	RGB	76.27	97.98	80.37	96.75
	Texture	61.05	92.98	65.48	94.00

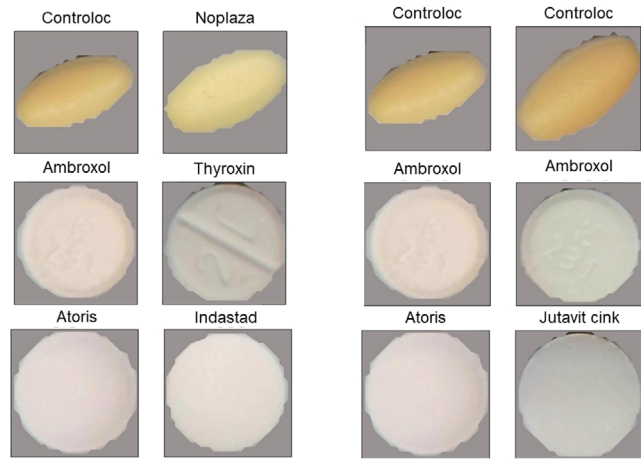


FIGURE 7 In the image pairs on the left, the images of the same types of pills are classified without any high-level cues. Conversely, in the image pairs on the right, the classification is performed using high-level cues resulting in fewer mistakes (only in the last row).

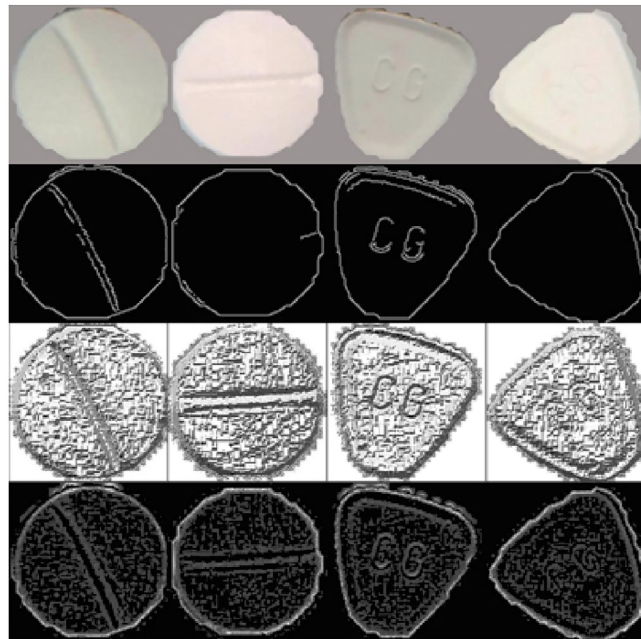


FIGURE 8 Illustration of different lightings on sub-streams by two pills (one pill in first two columns and other pill in third and fourth columns). First row: RGB images; second row: contours; third row: LBP; fourth row: texture.

TABLE 12 Inference time of certain modules in the proposed MSFN.

	Inference time [ms]	
	CPU	GPU
Segmentation	148.30	12.50
Stream - Contour	42.51	18.51
Stream - LBP	51.01	20.00
Stream - RGB	54.52	21.00
Stream - Texture	43.01	19.00
Fusion	7.25	2.00
Comparison	1.00	1.00
Total	347.60	94.02

6 | CONCLUSION

In this article, we introduced a new metric model and training mechanism for pill recognition. We believe metric learning methods are the most adequate approaches due to the possible few-shot application scenarios. While large backbone networks are well-proved for different image recognition tasks, there are two ways that we can improve the classification accuracy of models built upon them. First, our uniform, multi-stream approach is designed to efficiently capture medical pills' visual design intentions. Second, the metric learning model can be improved by injecting high-level (semantic) information about the presumed appearance of classes into the loss function. This way, the discriminating capability of the metric model increases. It is noteworthy that besides well-proven image feature descriptors (such as Fourier contour descriptors, encodings of scores and surface signs, colour descriptors), even free text description of the pills in the leaflets, processed with automatic NLP methods, can be utilised to improve the accuracy of the metric model. With the help of our proposed modifications and techniques, the new solution could outperform previous methods [6, 7, 9, 30] in Top-1 and Top-5 accuracy.

While the framework we proposed can be considered general and applicable to other classification problems, the applied feature extraction methods, datasets, and ablation studies are specific to the problem of pill recognition, which is a serious problem in health care. A different use-case on different visual objects would probably require different sub-streams. Since the

proposed dynamic triplet loss function can be used in many areas of metric learning, our approach could be applied in other fields of recognition (e.g. face recognition [8], human pose estimation [43], or industrial object recognition) where discriminating features can be numerically quantified based on reference quality images or semantic descriptions.

We have shown, with numerous tests, the high accuracy of our proposal and its readiness for different applications for pill verification. In future work, the model could also consider size information to reach the best possible results in unregulated conditions. The integration of an approach similar to [28], with the help of contextual information, could tackle the size problem.

AUTHOR CONTRIBUTIONS

Richárd Rádli: Methodology; software; validation; writing—reviewing and editing; conceptualization; data curation. **Zsolt Vörösházi:** Visualization; writing—reviewing and editing; data curation; methodology. **László Czúni:** Methodology; supervision; writing—reviewing and editing; conceptualization.

ACKNOWLEDGEMENTS

This work has been partly supported by the 2020-1.1.2-PIACI-KFI-2021-00296 and the TKP2021-NVA-10 project of the National Research, Development and Innovation Fund. We also acknowledge the financial support of the Hungarian Scientific Research Fund Grant OTKA K-135729. We are grateful to the NVIDIA corporation for supporting our research with GPUs obtained by the NVIDIA Hardware Grant Program. Last but not least, we would like to thank József Bene for his support in creating the OGYEIV2 dataset.

CONFLICT OF INTEREST STATEMENT

The authors declare no potential conflict of interests.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in OGYEIV2 dataset at <https://www.kaggle.com/datasets/richardradli/ogyeiv2>.

ORCID

Zsolt Vörösházi  <https://orcid.org/0009-0004-3032-8784>
László Czúni  <https://orcid.org/0000-0001-7667-9513>

REFERENCES

- Cronenwett, L.R., Bootman, J.L., Wolcott, J., Aspden, P.: Preventing Medication Errors. National Academies Press, Washington, DC (2007)
- Beso, A., Franklin, B.D., Barber, N.: The frequency and potential causes of dispensing errors in a hospital pharmacy. *Pharm. World Sci.* 27, 182–190 (2005)
- James, K.L., Barlow, D., Burfield, R., Hiom, S., Roberts, D., Whittlesea, C.: Unprevented or prevented dispensing incidents: which outcome to use in dispensing error research? *Int. J. Pharm. Pract.* 19(1), 36–50 (2011)
- Yaniv, Z., et al.: The National Library of Medicine pill image recognition challenge: An initial report. In: 2016 IEEE Applied Imagery Pattern Recognition Workshop (AIPR), pp. 1–9. IEEE, Piscataway (2016)
- Zeng, X., Cao, K., Zhang, M.: Mobiledeeppill: A small-footprint mobile deep learning system for recognizing unconstrained pill images. In: Proceedings of the 15th Annual International Conference on Mobile Systems, Applications, and Services, pp. 56–67. ACM, New York (2017)
- Ling, S., Pastor, A., Li, J., Che, Z., Wang, J., Kim, J., Callet, P.L.: Few-shot pill recognition. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 9789–9798. IEEE, Piscataway (2020)
- Pastor, A., Ling, S., Kim, J., Le Callet, P.: Case study: Few-shot pill recognition: How to train an AI model to recognize a new category of pill from only a few samples like humans? In: Meta Learning with Medical Imaging and Health Informatics Applications, pp. 275–299. Academic Press, Washington, DC (2023)
- Schroff, F., Kalenichenko, D., Philbin, J.: Facenet: A unified embedding for face recognition and clustering. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 815–823. IEEE, Piscataway (2015)
- Rádli, R., Vörösházi, Z., Czúni, L.: Multi-stream pill recognition with attention. In: 2023 IEEE 12th International Conference on Intelligent Data Acquisition and Advanced Computing Systems: Technology and Applications (IDAACS), vol. 1, pp. 942–946. IEEE, Piscataway (2023)
- Ou, Y.Y., Tsai, A.C., Wang, J.F., Lin, J.: Automatic drug pills detection based on convolution neural network. In: 2018 International Conference on Orange Technologies (ICOT), pp. 1–4. IEEE, Piscataway (2018)
- He, K., Zhang, X., Ren, S., Sun, J.: Deep residual learning for image recognition. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 770–778. IEEE, Piscataway (2016)
- Chollet, F.: Xception: Deep learning with depthwise separable convolutions. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 1251–1258. IEEE, Piscataway (2017)
- Kwon, H.J., Kim, H.G., Lee, S.H.: Pill detection model for medicine inspection based on deep learning. *Chemosensors* 10(1), 4 (2021)
- He, K., Gkioxari, G., Dollár, P., Girshick, R.: Mask R-CNN. In: Proceedings of the IEEE International Conference on Computer Vision, pp. 2961–2969. IEEE, Piscataway (2017)
- Tan, L., Huangfu, T., Wu, L., Chen, W.: Comparison of RetinaNet, SSD, and YOLOv3 for real-time pill identification. *BMC Med. Inf. Decis. Making* 21, 1–11 (2021)
- Redmon, J., Farhadi, A.: YOLOv3: An incremental improvement. *arXiv preprint, arXiv:1804.02767* (2018)
- Lin, T.Y., Goyal, P., Girshick, R., He, K., Dollár, P.: Focal loss for dense object detection. In: Proceedings of the IEEE International Conference on Computer Vision, pp. 2980–2988. IEEE, Piscataway (2017)
- Liu, W., Anguelov, D., Erhan, D., Szegedy, C., Reed, S., Fu, C.-Y., Berg, A.C.: SSD: Single shot multibox detector. In: Computer Vision—ECCV 2016: 14th European Conference, Proceedings, Part I 14, pp. 21–37. Springer, Cham (2016)
- Pfaff, B., Gepperth, A., Abdullah, S., Kilian, A.: Catastrophic forgetting: still a problem for DNNs. In: Artificial Neural Networks and Machine Learning—ICANN 2018: 27th International Conference on Artificial Neural Networks, Proceedings, Part I 27, pp. 487–497. Springer, Cham (2018)
- Nagy, A.M., Czúni, L.: Classification and fast few-shot learning of steel surface defects with randomized network. *Appl. Sci.* 12(8), 3967 (2022)
- Doan, H.G., Luong, H.Q., Ha, T.O., Pham, T.T.T.: An efficient strategy for catastrophic forgetting reduction in incremental learning. *Electronics* 12(10), 2265 (2023)
- Busta, M., Neumann, L., Matas, J.: Deep textspotter: An end-to-end trainable scene text localization and recognition framework. In: Proceedings of the IEEE International Conference on Computer Vision, pp. 2204–2212. IEEE, Piscataway (2017)
- Ojala, T., Pietikainen, M., Harwood, D.: Performance evaluation of texture measures with classification based on Kullback discrimination of distributions. In: Proceedings of 12th International Conference on Pattern Recognition, vol. 1, pp. 582–585. IEEE, Piscataway (1994)
- Liu, L., Zhang, H., Feng, A., Wan, X., Guo, J.: Simplified local binary pattern descriptor for character recognition of vehicle license plate. In: 2010 Seventh International Conference on Computer Graphics, Imaging and Visualization, pp. 157–161. IEEE, Piscataway (2010)

25. Hassan, T., Khan, H.A.: Handwritten bangla numeral recognition using local binary pattern. In: 2015 International Conference on Electrical Engineering and Information Communication Technology (ICEEICT), pp. 1–4. IEEE, Piscataway (2015)
26. Heo, J., Kang, Y., Lee, S., Jeong, D.H., Kim, K.M.: An accurate deep learning-based system for automatic pill identification: Model development and validation. *J. Med. Internet Res.* 25, e41043 (2023)
27. Nguyen, A.D., Nguyen, T.D., Pham, H.H., Nguyen, T.H., Nguyen, P.L.: Image-based contextual pill recognition with medical knowledge graph assistance. In: Asian Conference on Intelligent Information and Database Systems, pp. 354–369. Springer, Singapore (2022)
28. Nguyen, A.D., Pham, H.H., Trung, H.T., Nguyen, Q.V.H., Truong, T.N., Nguyen, P.L.: High accurate and explainable multi-pill detection framework with graph neural network-assisted multimodal data fusion. *PLoS One* 18(9), e0291865 (2023)
29. Zhang, J., Liu, L., Gao, K., Hu, D.: A forward and backward compatible framework for few-shot class-incremental pill recognition. *arXiv preprint, arXiv:2304.11959* (2023)
30. Rádli, R., Vörösházi, Z., Czúni, L.: Word and image embeddings in pill recognition. In: Proceedings of the 19th International Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications (VISIGRAPP 2024), vol. 3, pp. 729–736. SciTePress, Setúbal, Portugal (2024)
31. VAIPE: VAIPE-Pill—A large-scale, annotated benchmark dataset for visual pill identification (2008). <https://vaipe.org/>. Accessed 1 July 2024
32. Rádli, R., Bene, J., Vörösházi, Z.: OGYEIV2 (2023). <https://www.kaggle.com/dsv/8417535>. Accessed 1 July 2024.
33. Jocher, G., Chaurasia, A., Qiu, J.: Ultralytics YOLOv8 (2023). <https://github.com/ultralytics/ultralytics>. Accessed 1 July 2024.
34. Cimpoi, M., Maji, S., Kokkinos, I., Mohamed, S., Vedaldi, A.: Describing textures in the wild. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 3606–3613. IEEE, Piscataway (2014)
35. Ronneberger, O., Fischer, P., Brox, T.: U-net: Convolutional networks for biomedical image segmentation. In: Medical Image Computing and Computer-assisted Intervention (MICCAI 2015): 18th International Conference, vol. Part III 18, pp. 234–241. Springer, Cham (2015)
36. Vaswani, A., et al.: Attention is all you need. In: Advances in Neural Information Processing Systems, vol. 30. Curran Associates, Red Hook, NY (2017)
37. Tan, M., Le, Q.: EfficientNetv2: Smaller models and faster training. In: International Conference on Machine Learning, pp. 10096–10106. PMLR, New York (2021)
38. Orosz, G., Szabó, G., Berkecz, P., Szántó, Z., Farkas, R.: Advancing Hungarian text processing with HuSpaCy: Efficient and accurate NLP pipelines. In: Ekšteín, K., Pártl, F., Konopík, M. (eds.) Text, Speech, and Dialogue, pp. 58–69. Springer Nature, Cham (2023)
39. Van der Maaten, L., Hinton, G.: Visualizing data using t-SNE. *J. Mach. Learn. Res.* 9(11), 2579–2605 (2008)
40. Standard, C., et al.: Colorimetry-part 4: CIE 1976 L* a* b* colour space. International Standard 2019–06 (2007)
41. Kuhl, F.P., Giardina, C.R.: Elliptic Fourier features of a closed contour. *Comp. Graph. Image Process.* 18(3), 236–258 (1982)
42. Chopra, S., Hadsell, R., LeCun, Y.: Learning a similarity metric discriminatively, with application to face verification. In: 2005 IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR'05), vol. 1, pp. 539–546. IEEE, Piscataway (2005)
43. Kim, S., Seo, M., Laptev, I., Cho, M., Kwak, S.: Deep metric learning beyond binary supervision. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 2288–2297. IEEE, Piscataway (2019)
44. Liaw, R., Liang, E., Nishihara, R., Moritz, P., Gonzalez, J.E., Stoica, I.: Tune: A research platform for distributed model selection and training. *arXiv preprint, arXiv:1807.05118* (2018)

How to cite this article: Rádli, R., Vörösházi, Z., Czúni, L.: Metric-based pill recognition with the help of textual and visual cues. *IET Image Process.* 18, 4623–4638 (2024). <https://doi.org/10.1049/ipr2.13273>